

数学物理方法作业

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第十七章

1.(1)

$$\begin{aligned}\int_0^1 \sqrt{1-x} \sin(a\sqrt{x}) dx &= \int_0^{\frac{\pi}{2}} \cos \theta \sin(a \sin \theta) \cdot (2 \sin \theta \cos \theta d\theta) \\&= \int_0^{\frac{\pi}{2}} \cos \theta \sin(a \sin \theta) \sin 2\theta d\theta \\&= -\frac{1}{a} \int_0^{\frac{\pi}{2}} \sin 2\theta [-\sin(a \sin \theta) a \cos \theta d\theta] \\&= -\frac{1}{a} \int_0^{\frac{\pi}{2}} \sin 2\theta d[\cos(a \sin \theta)] \\&= \frac{1}{a} \cos(a \sin \theta) \sin 2\theta \Big|_0^{\frac{\pi}{2}} - \frac{1}{a} \int_0^{\frac{\pi}{2}} \sin 2\theta d[\cos(a \sin \theta)] \\&= \frac{1}{a} \int_0^{\frac{\pi}{2}} \cos(a \sin \theta) d(\sin 2\theta) \\&= \frac{2}{a} \int_0^{\frac{\pi}{2}} \cos(a \sin \theta) \cos 2\theta d\theta \\&= \frac{2}{a} \int_0^{\frac{\pi}{2}} [\cos(a \sin \theta) \cos 2\theta + \sin(a \sin \theta) \sin 2\theta] d\theta \\&= \frac{1}{a} \int_0^{\pi} [\cos(a \sin \theta) \cos 2\theta + \sin(a \sin \theta) \sin 2\theta] d\theta \\&= \frac{1}{a} \int_0^{\pi} \cos(a \sin 2\theta - 2\theta) d\theta \\&= \frac{\pi}{a} \cdot \frac{1}{\pi} \int_0^{\pi} \cos(a \sin 2\theta - 2\theta) d\theta \\&= \frac{\pi}{a} J_2(a)\end{aligned}$$

1.(3)

$$\begin{aligned}
& \int_0^\infty e^{-ax} J_\nu(bx) x^{\nu+1} dx \\
&= \int_0^\infty e^{-ax} \sum_{k=0}^\infty \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{bx}{2}\right)^{2k+\nu} x^{\nu+1} dx \\
&= \int_0^\infty e^{-ax} \sum_{k=0}^\infty \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{b}{2}\right)^{2k+\nu} x^{2k+2\nu+1} dx \\
&= \sum_{k=0}^\infty \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{b}{2}\right)^{2k+\nu} \int_0^\infty e^{-ax} x^{2k+2\nu+1} dx \\
&= \sum_{k=0}^\infty \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \frac{b^{2k+\nu}}{2^{2k+\nu} a^{2k+2\nu+2}} \int_0^\infty e^{-ax} (ax)^{2k+2\nu+1} d(ax) \\
&= \sum_{k=0}^\infty \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \frac{b^{2k+\nu}}{2^{2k+\nu} a^{2k+2\nu+2}} \Gamma(2k + 2\nu + 2) \\
&= \sum_{k=0}^\infty \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \frac{b^{2k+\nu}}{2^{2k+\nu} a^{2k+2\nu+2}} \cdot \frac{2^{2k+2\nu+1}}{\sqrt{\pi}} \Gamma(k + \nu + 1) \Gamma(k + \nu + \frac{3}{2}) \\
&= \sum_{k=0}^\infty \frac{(-1)^k}{k!} \frac{b^{2k+\nu}}{a^{2k+2\nu+2}} \cdot \frac{2^{\nu+1}}{\sqrt{\pi}} \Gamma\left(k + \nu + \frac{3}{2}\right) \\
&= \frac{2^{\nu+1} b^\nu}{\sqrt{\pi} a^{2\nu+2}} \sum_{k=0}^\infty \frac{(-1)^k}{k!} \left(\frac{b}{a}\right)^{2k} \Gamma\left(k + \nu + \frac{3}{2}\right) \\
&= \frac{2^{\nu+1} b^\nu}{\sqrt{\pi} a^{2\nu+2}} \sum_{k=0}^\infty \frac{(-1)^k}{k!} \left(k + \nu + \frac{1}{2}\right) \cdots \left(1 + \nu + \frac{1}{2}\right) \left(\frac{b}{a}\right)^{2k} \Gamma\left(\nu + \frac{3}{2}\right) \\
&= \frac{2^{\nu+1} b^\nu}{\sqrt{\pi} a^{2\nu+2}} \Gamma\left(\nu + \frac{3}{2}\right) \sum_{k=0}^\infty \frac{1}{k!} \left(-k - \nu - \frac{1}{2}\right) \cdots \left(-1 - \nu - \frac{1}{2}\right) \left(\frac{b}{a}\right)^{2k} \\
&= \frac{2^{\nu+1} b^\nu}{\sqrt{\pi} a^{2\nu+2}} \Gamma\left(\nu + \frac{3}{2}\right) \left[1 + \left(\frac{b}{a}\right)^2\right]^{-\nu-\frac{3}{2}} \\
&= \frac{2^{\nu+1} a b^\nu}{\sqrt{\pi} a^{2\nu+3}} \Gamma\left(\nu + \frac{3}{2}\right) \left[1 + \left(\frac{b}{a}\right)^2\right]^{-\nu-\frac{3}{2}} \\
&= \frac{2^{\nu+1} a b^\nu}{\sqrt{\pi}} \Gamma\left(\nu + \frac{3}{2}\right) (a^2 + b^2)^{-\nu-\frac{3}{2}} \\
&= \frac{2^{\nu+1} a b^\nu}{\sqrt{\pi}} \Gamma\left(\nu + \frac{3}{2}\right) (a^2 + b^2)^{-\nu-\frac{3}{2}} \\
&= \frac{2a(2b)^\nu}{\sqrt{\pi}} \frac{\Gamma\left(\nu + \frac{3}{2}\right)}{(a^2 + b^2)^{\nu+\frac{3}{2}}}
\end{aligned}$$

2.(1) 由于:

$$e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = \sum_{n=-\infty}^{\infty} J_n(x)t^n$$

$$\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$$

令 $t = i$ 得:

$$\begin{aligned}\cos x &= \frac{1}{2}(e^{ix} + e^{-ix}) \\ &= \frac{1}{2}\left(e^{\frac{x}{2}\left(i-\frac{1}{i}\right)} + e^{\frac{x}{2}\left(-i+\frac{1}{i}\right)}\right) \\ &= \frac{1}{2}\left[\sum_{n=-\infty}^{\infty} J_n(x)i^n + \sum_{n=-\infty}^{\infty} J_n(x)(-i)^n\right] \\ &= \sum_{n=-\infty}^{\infty} J_n(x) \cdot \frac{i^n - (-i)^n}{2} \\ &= \sum_{n=-\infty}^{\infty} (-1)^n J_{2n}(x)\end{aligned}$$

由

$$J_{-n}(x) = (-1)^n J_n(x)$$

得

$$J_{-2n}(x) = (-1)^{2n} J_{2n}(x) = J_{2n}(x)$$

于是:

$$\begin{aligned}\cos x &= \sum_{n=-\infty}^{\infty} (-1)^n J_{2n}(x) \\ &= J_0(x) + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(x)\end{aligned}$$

同理, 由于:

$$\sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$$

令 $t = i$ 得:

$$\begin{aligned}\sin x &= \frac{1}{2}(e^{ix} - e^{-ix}) \\ &= \frac{1}{2i}\left(e^{\frac{x}{2}\left(i-\frac{1}{i}\right)} - e^{\frac{x}{2}\left(-i+\frac{1}{i}\right)}\right) \\ &= \frac{1}{2i}\left[\sum_{n=-\infty}^{\infty} J_n(x)i^n - \sum_{n=-\infty}^{\infty} J_n(x)(-i)^n\right] \\ &= \sum_{n=-\infty}^{\infty} J_n(x) \cdot \frac{i^n - (-i)^n}{2i}\end{aligned}$$

$$= \sum_{n=0}^{\infty} (-1)^n J_{2n+1}(x) - \sum_{n=-\infty}^0 (-1)^n J_{-2n-1}(x)$$

由

$$J_{-n}(x) = (-1)^n J_n(x)$$

得

$$J_{-2n-1}(x) = (-1)^{2n+1} J_{2n+1}(x) = -J_{2n+1}(x)$$

于是:

$$\begin{aligned} \sin x &= \sum_{n=0}^{\infty} (-1)^n J_{2n+1}(x) - \sum_{n=-\infty}^0 (-1)^n J_{-2n-1}(x) \\ &= 2 \sum_{n=0}^{\infty} (-1)^n J_{2n+1}(x) \end{aligned}$$

2.(3) 由于:

$$e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = \sum_{n=-\infty}^{\infty} J_n(x) t^n$$

令 $t = 1$ 得:

$$1 = \sum_{n=-\infty}^{\infty} J_n(x) = J_0(x) + 2 \sum_{n=1}^{\infty} J_{2n}(x)$$

由于:

$$J'_{\nu}(x) = \frac{1}{2} [J_{\nu-1}(x) - J_{\nu+1}(x)]$$

上式两边对 x 求导得:

$$J''_{\nu}(x) = \frac{1}{2} [J'_{\nu-1}(x) - J'_{\nu+1}(x)]$$

代入:

$$J'_{\nu+1}(x) = \frac{1}{2} [J_{\nu}(x) - J_{\nu+2}(x)]$$

$$J'_{\nu-1}(x) = \frac{1}{2} [J_{\nu-2}(x) - J_{\nu}(x)]$$

得:

$$J''_{\nu}(x) = \frac{1}{4} J_{\nu+2}(x) - \frac{1}{2} J_{\nu}(x) + \frac{1}{4} J_{\nu-2}(x)$$

另一方面:

$$\begin{aligned} & \frac{d^2}{dx^2} \left[2 \sum_{n=1}^{\infty} (2n)^2 J_{2n}(x) \right] \\ &= 2 \sum_{n=1}^{\infty} (2n)^2 J''_{2n}(x) \end{aligned}$$

$$\begin{aligned}
&= 2 \sum_{n=1}^{\infty} (2n)^2 \left[\frac{1}{4} J_{2n+2}(x) - \frac{1}{2} J_{2n}(x) + \frac{1}{4} J_{2n-2}(x) \right] \\
&= 2 \left\{ J_0(x) + 2J_2(x) + \sum_{n=2}^{\infty} [(n-1)^2 - 2n^2 + (n+1)^2] J_{2n}(x) \right\} \\
&= 2 \left[J_0(x) + 2J_2(x) + \sum_{n=2}^{\infty} 2J_{2n}(x) \right] \\
&= 2 \left[J_0(x) + 2 \sum_{n=1}^{\infty} J_{2n}(x) \right] \\
&= 2
\end{aligned}$$

即级数

$$y = 2 \sum_{n=1}^{\infty} (2n)^2 J_{2n}(x)$$

满足微分方程:

$$\frac{d^2 y}{dx^2} = 2$$

该方程的通解为:

$$y = x^2 + px + q$$

又因为 y 仅含偶数阶 Bessel 函数, 为偶函数, 则 $p = 0$; 且 $x = 0$ 时 $y = 0$, 则 $q = 0$ 。
故解得:

$$x^2 = 2 \sum_{n=1}^{\infty} (2n)^2 J_{2n}(x)$$

3.(1)

$$\begin{aligned}
\int_0^x x^{-n} J_{n+1}(x) dx &= - \int_0^x d[x^{-n} J_n(x)] \\
&= -x^{-n} J_n(x) \Big|_0^x \\
&= -x^{-n} J_n(x) + \lim_{x \rightarrow 0} \frac{J_n(x)}{x^n}
\end{aligned}$$

代入 Bessel 函数在 $x \rightarrow 0$ 时的渐进表达式:

$$J_n(x) = \frac{1}{n!} \left(\frac{x}{2} \right)^n + O(x^{n+2})$$

得:

$$\begin{aligned}
\int_0^x x^{-n} J_{n+1}(x) dx &= -x^{-n} J_n(x) + \lim_{x \rightarrow 0} \frac{J_n(x)}{x^n} \\
&= -x^{-n} J_n(x) + \lim_{x \rightarrow 0} \left[\frac{1}{n!} \left(\frac{1}{2} \right)^n + O(x^2) \right]
\end{aligned}$$

$$= -x^{-n} J_n(x) + \frac{1}{2^n n!}$$

3.(3)

$$\begin{aligned}
& \int_0^t J_0(\sqrt{x(t-x)}) dx \\
&= \int_0^t \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left[\frac{\sqrt{x(t-x)}}{2} \right]^{2k} dx \\
&= \int_0^t \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left[\frac{x(t-x)}{4} \right]^k dx \\
&= \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2 4^k} \int_0^t x^k (t-x)^k dx \\
&= \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2 4^k} \cdot t^{2k+1} \int_0^1 \left(\frac{x}{t} \right)^k \left[1 - \left(\frac{x}{t} \right) \right]^k d\left(\frac{x}{t} \right) \\
&= \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2 4^k} \cdot t^{2k+1} B(k+1, k+1) \\
&= \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2 4^k} \cdot t^{2k+1} \frac{(k!)^2}{(2k+1)!} \\
&= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \frac{t^{2k+1}}{2^{2k}} \\
&= 2 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \left(\frac{t}{2} \right)^{2k+1} \\
&= 2 \sin \frac{t}{2}
\end{aligned}$$

4. 膜振动问题与 θ 无关, 得 r 方向方程:

$$\frac{\partial^2 u}{\partial t^2} - \frac{c^2}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) = 0$$

其中 c 为波速, 边界条件和初始条件为:

$$\begin{aligned}
& |u|_{r=0} < \infty, \quad u|_{r=R} = 0 \\
& u|_{t=0} = A \left(1 - \frac{r^2}{R^2} \right), \quad \frac{\partial u}{\partial t} \Big|_{t=0} = 0
\end{aligned}$$

分离变量:

$$u(r, t) = D(r)T(t)$$

得到:

$$\begin{aligned} T''(t) + c^2 k^2 T(t) &= 0, & \frac{1}{r} \frac{d}{dr} \left(r \frac{dD}{dr} \right) + k^2 D(r) &= 0 \\ |D(0)| &< \infty, & D(R) &= 0 \\ D|_{t=0} &= A \left(1 - \frac{r^2}{R^2} \right), & \frac{\partial D}{\partial t} \Big|_{t=0} &= 0 \end{aligned}$$

令 $x = kr$, $y(x) = D(r)$ 得:

$$\frac{1}{x} \frac{d}{dx} \left(x \frac{dy}{dx} \right) + y(x) = 0$$

这是一个零阶 Bessel 方程, 通解为:

$$D(r) = PJ_0(kr) + QN_0(kr)$$

由 $|D(0)| < \infty$ 得 $Q = 0$, 则 $P \neq 0$, 取 $P = 1$ 。

由 $D(R) = 0$ 得 $J_0(kR) = 0$, 设 μ_i 是 $J_0(x)$ 从小到大的第 i 个正零点, 则 $kR = \mu_i$ 。

本征值和本征函数分别为:

$$k_i^2 = \left(\frac{\mu_i}{R} \right)^2, \quad D_i(r) = J_0 \left(\frac{\mu_i r}{R} \right)$$

代入关于 T 的方程得:

$$T_i(t) = A_i \cos \frac{c\mu_i}{R} t + B_i \sin \frac{c\mu_i}{R} t$$

一般解为:

$$u(r, t) = \sum_{i=1}^{\infty} J_0 \left(\frac{\mu_i r}{R} \right) \left[A_i \cos \frac{c\mu_i}{R} t + B_i \sin \frac{c\mu_i}{R} t \right]$$

由 $\frac{\partial u}{\partial t} \Big|_{t=0} = 0$ 得: $B_i = 0$; 由 $u|_{t=0} = A \left(1 - \frac{r^2}{R^2} \right)$ 得:

$$u(r, t)|_{t=0} = A \left(1 - \frac{r^2}{R^2} \right) = \sum_{i=1}^{\infty} A_i J_0 \left(\frac{\mu_i r}{R} \right)$$

由本征函数的正交性叠加系数得:

$$A_i = \frac{1}{\left\| J_0 \left(\frac{\mu_i r}{R} \right) \right\|^2} \int_0^R A \left(1 - \frac{r^2}{R^2} \right) J_0 \left(\frac{\mu_i r}{R} \right) r \, dr$$

令 $x = \frac{r}{R}$ 得:

$$\begin{aligned} A_i &= \frac{1}{\left\| J_0 \left(\frac{\mu_i r}{R} \right) \right\|^2} \int_0^R A \left(1 - \frac{r^2}{R^2} \right) J_0 \left(\frac{\mu_i r}{R} \right) r \, dr \\ &= \frac{AR^2}{\left\| J_0 \left(\frac{\mu_i r}{R} \right) \right\|^2} \int_0^1 (1 - x^2) J_0(\mu_i x) x \, dx \end{aligned}$$

$$\begin{aligned}
&= \frac{AR^2}{\left\| J_0\left(\frac{\mu_i r}{R}\right) \right\|^2} \cdot \frac{2J_2(\mu_i)}{\mu_i^2} \\
&= \frac{2AR^2}{R^2 J_1^2(\mu_i)} \cdot \frac{4J_1(\mu_i)}{\mu_i^3} \\
&= \frac{8A}{\mu_i^3 J_1(\mu_i)}
\end{aligned}$$

故原问题的解为:

$$u(r, t) = \sum_{i=1}^{\infty} \frac{8A}{\mu_i^3 J_1(\mu_i)} J_0\left(\frac{\mu_i r}{R}\right) \cos \frac{c\mu_i}{R} t$$

5. 分离变量:

$$u = R(r)\Phi(\phi)T(t)$$

得:

$$\begin{aligned}
\frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \left(k^2 - \frac{\lambda}{r^2} \right) R &= 0, \quad |R(0)| < \infty, \quad R(a) = 0 \\
\Phi''(\phi) + \lambda \Phi(\phi) &= 0, \quad \Phi(0) = \Phi(2\pi), \quad \Phi'(0) = \Phi'(2\pi) \\
T'(t) + \kappa k^2 T &= 0
\end{aligned}$$

对于 ϕ 的本征值问题, 采用复本征函数:

$$\lambda = m^2, \quad m \in \mathbb{Z}$$

$$\Phi_m(\phi) = e^{im\phi}$$

对于 r 的本征值问题, 令 $x = kr$, $y(x) = R(r)$, 则方程化为 Bessel 方程, 其通解为:

$$R(r) = C J_m(kr) + D N_m(kr)$$

由 $|R(0)| < \infty$ 得 $D = 0$, 则 $C \neq 0$, 取 $C = 1$ 。

由 $R(a) = 0$ 得 $J_m(kR) = 0$, 设 $\mu_j^{(m)}$ 是 $J_m(x)$ 从小到大的第 j 个正零点, 则 $ka = \mu_j^{(m)}$ 。

本征值和本征函数分别为:

$$k_{mj}^2 = \left(\frac{\mu_j^{(m)}}{a} \right)^2, \quad R_{mj}(r) = J_0\left(\frac{\mu_j^{(m)} r}{R}\right)$$

代入关于 T 的方程得:

$$T_j(t) = C_{mj} e^{-\kappa k_{mj}^2 t}$$

一般解为:

$$u(r, \phi, t) = \sum_{j=1}^{\infty} \sum_{m=-\infty}^{\infty} J_m\left(\frac{\mu_j^{(m)}}{a}\right) e^{im\phi} C_{mj} e^{-\kappa k_{mj}^2 t}$$

由 $u|_{t=0} = u_0 \sin(2\phi)$ 得:

$$u_0 \sin(2\phi) = \sum_{j=1}^{\infty} \sum_{m=-\infty}^{\infty} J_m \left(\frac{\mu_j^{(m)}}{a} \right) e^{im\phi} C_{mj}$$

由本征函数的正交性叠加系数得:

$$\begin{aligned} C_{mj} &= \frac{1}{2\pi \left\| J_m \left(\frac{\mu_j^{(m)} r}{a} \right) \right\|^2} \int_0^a \int_0^{2\pi} J_m \left(\frac{\mu_j^{(m)} r}{a} \right) u_0 \sin(2\phi) e^{im\phi} r dr d\phi \\ &= \frac{1}{2\pi \left\| J_m \left(\frac{\mu_j^{(m)} r}{a} \right) \right\|^2} \int_0^a J_m \left(\frac{\mu_j^{(m)} r}{a} \right) r dr \int_0^{2\pi} u_0 \sin(2\phi) e^{im\phi} d\phi \end{aligned}$$

注意到:

$$\begin{aligned} \int_0^{2\pi} u_0 \sin(2\phi) e^{im\phi} d\phi &= \int_0^{2\pi} \frac{u_0}{2i} [e^{i(m+2)\phi} - e^{i(m-2)\phi}] d\phi \\ &= \begin{cases} \pm \pi i u_0, & m = \pm 2 \\ 0, & m \neq \pm 2 \end{cases} \end{aligned}$$

又 $J_2(x) = J_{-2}(x)$, 本征值只能取 $m = 2$ 。

下面省略 C 下标的 2 和 μ_j 上标的 (2), 此时 C_j 的取值为:

$$\begin{aligned} C_j &= \frac{\pi i u_0}{2\pi \left\| J_2 \left(\frac{\mu_j r}{a} \right) \right\|^2} \int_0^a J_2 \left(\frac{\mu_j r}{a} \right) r dr \\ &= \frac{2i u_0}{\mu_j^4 J_0^2(\mu_j)} [4 - (\mu_j^2 + 4) J_0(\mu_j)] \end{aligned}$$

故原问题的解为:

$$u(r, \phi, t) = 4u_0 \sin(2\phi) \sum_{j=1}^{\infty} \frac{4 - (\mu_j^2 + 4) J_0(\mu_j)}{\mu_j^4 J_0^2(\mu_j)} J_2 \left(\frac{\mu_j r}{a} \right) e^{-\kappa \left(\frac{\mu_j}{a} \right)^2 t}$$

其中 μ_j 是 $J_2(x)$ 从小到大的第 j 个正零点。

8.(1) 膜振动问题与 θ 无关, 得 r 方向方程:

$$\frac{\partial^2 u}{\partial t^2} - \frac{c^2}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) = A \sin \omega t$$

对应的齐次方程为:

$$\frac{\partial^2 u}{\partial t^2} - \frac{c^2}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) = 0$$

其中 c 为波速, 边界条件和初始条件为:

$$\begin{aligned} |u|_{r=0} < \infty, \quad u|_{r=R} &= 0 \\ u|_{t=0} &= 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0 \end{aligned}$$

分离变量:

$$u(r, t) = D(r)T(t)$$

得到:

$$\begin{aligned} T''(t) + c^2 k^2 T(t) &= 0, \quad \frac{1}{r} \frac{d}{dr} \left(r \frac{dD}{dr} \right) + k^2 D(r) = 0 \\ |D(0)| < \infty, \quad D(R) &= 0 \\ D|_{t=0} &= 0, \quad \left. \frac{\partial D}{\partial t} \right|_{t=0} = 0 \end{aligned}$$

令 $x = kr$, $y(x) = D(r)$ 得:

$$\frac{1}{x} \frac{d}{dx} \left(x \frac{dy}{dx} \right) + y(x) = 0$$

这是一个零阶 Bessel 方程, 通解为:

$$D(r) = PJ_0(kr) + QN_0(kr)$$

由 $|D(0)| < \infty$ 得 $Q = 0$, 则 $P \neq 0$, 取 $P = 1$ 。

由 $D(R) = 0$ 得 $J_0(kR) = 0$, 设 μ_i 是 $J_0(x)$ 从小到大的第 i 个正零点, 则 $kR = \mu_i$ 。

本征值和本征函数分别为:

$$k_i^2 = \left(\frac{\mu_i}{R} \right)^2, \quad D_i(r) = J_0 \left(\frac{\mu_i r}{R} \right)$$

一般解为:

$$u(r, t) = \sum_{i=1}^{\infty} J_0 \left(\frac{\mu_i r}{R} \right) T_i(t)$$

将 $u(r, t)$ 代入

$$\frac{\partial^2 u}{\partial t^2} - \frac{c^2}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) = A \sin \omega t$$

得到:

$$\begin{aligned} \sum_{i=1}^{\infty} J_0 \left(\frac{\mu_i r}{R} \right) T_i''(t) + \sum_{i=1}^{\infty} \left(\frac{\mu_i c}{R} \right)^2 J_0 \left(\frac{\mu_i r}{R} \right) T_i(t) &= \sum_{i=1}^{\infty} J_0 \left(\frac{\mu_i r}{R} \right) A_i \sin \omega t \\ T_i(0) &= 0, \quad T_i'(0) = 0 \end{aligned}$$

即将非齐次项按相应齐次问题本征函数展开, 展开系数 A_i 满足:

$$\sum_{i=1}^{\infty} A_i J_0 \left(\frac{\mu_i r}{R} \right) = A$$

解得：

$$A_i = \frac{2A}{\mu_i J_1(\mu_i)}$$

方程化为：

$$T_i''(t) + \left(\frac{\mu_i c}{R}\right)^2 T_i(t) = \frac{2A}{\mu_i J_1(\mu_i)} \sin \omega t$$

再解出：

$$T_i(t) = \frac{2AR^2}{\mu_i J_1(\mu_i) [(\mu_i c)^2 - (\omega R)^2]} \left[\sin \omega t - \frac{\omega R}{\mu_i c} \sin \left(\frac{\mu_i c}{R} t \right) \right]$$

于是原问题的解为：

$$u(r, t) = \sum_{i=1}^{\infty} \frac{2AR^2}{\mu_i J_1(\mu_i) [(\mu_i c)^2 - (\omega R)^2]} \left[\sin \omega t - \frac{\omega R}{\mu_i c} \sin \left(\frac{\mu_i c}{R} t \right) \right] J_0 \left(\frac{\mu_i r}{R} \right)$$

发生共振时： $\omega_0 = \frac{\mu_i c}{R}$ 。

取 $\omega \rightarrow \omega_0$ 的极限，由 L'Hospital 法则得此时 $u(r, t)$ 的解为：

$$\begin{aligned} u(r, t) = & -\frac{AR}{c\mu_i^2 J_1(\mu_i)} J_0 \left(\frac{\mu_i r}{R} \right) \left[t \cos \left(\frac{\mu_i c}{R} t \right) - \frac{R}{\mu_i c} \sin \left(\frac{\mu_i c}{R} t \right) \right] \\ & + \sum_{i=1, i \neq j}^{\infty} \frac{2AR^2}{\mu_i^2 (\mu_i^2 - \mu_j^2) J_1(\mu_i)} \left[\mu_i \sin \left(\frac{\mu_i c}{R} t \right) - \mu_j \sin \left(\frac{\mu_j c}{R} t \right) \right] J_0 \left(\frac{\mu_i r}{R} \right) \end{aligned}$$

11. 温度分布与 θ 和 ϕ 均无关，得 r 方向的方程：

$$\frac{\partial u}{\partial t} - \frac{\kappa}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = 0$$

设 $u(r, t) = R(r)T(t)$ ，分离变量得：

$$\frac{1}{r} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + k^2 R(r) = 0$$

$$|R(0)| < \infty, \quad R(a) = 0, \quad R|_{t=0} = u_0$$

$$T'(t) + \kappa k^2 T(t) = 0$$

关于 r 的方程做代换 $x = kr$ ， $y(x) = R(r)$ 后可以化为零阶球 Bessel 方程，其解为：

$$y(x) = A j_0(x) + B n_0(x)$$

即：

$$R(r) = A j_0(kr) + B n_0(kr)$$

由 $|R(0)| < \infty$ 得 $B = 0$ ，则 $A \neq 0$ 。

由 $R(a) = 0$ 得 $j_0(ka) = 0$ ，则本征值和本征函数分别为：

$$k_n = \frac{\mu_n}{a}, \quad R_n(r) = j_0\left(\frac{\mu_n r}{a}\right)$$

其中 μ_n 是零阶球 Bessel 函数从小到大的第 n 个正零点。

代入关于 t 的方程，解得：

$$T_n(t) = e^{-\kappa\left(\frac{\mu_n}{a}\right)^2 t}$$

则一般解为：

$$u(r, t) = \sum_{i=1}^{\infty} A_n j_0\left(\frac{\mu_n r}{a}\right) e^{-\kappa\left(\frac{\mu_n}{a}\right)^2 t}$$

由 $u|_{t=0}$ 叠加系数：

$$u_0 = \sum_{i=n}^{\infty} A_n j_0\left(\frac{\mu_n r}{a}\right)$$

作代换 $x_n = k_n r = \frac{\mu_n r}{a}$ ，注意到 $j_0(x) = \frac{\sin x}{x}$ ，此时 $\mu_n = n\pi$ ， $x_n = \frac{n\pi r}{a}$ ，于是：

$$u_0 = \sum_{i=n}^{\infty} A_n j_0(x_n)$$

解得系数：

$$A_n = \frac{1}{\|j_0(x_n)\|^2} \int_0^a u_0 j_0(x_n) x_n^2 dx_n$$

得：

$$\begin{aligned} \|j_0(x_n)\|^2 &= \int_0^{n\pi} \left(\frac{\sin x_n}{x_n}\right)^2 x_n^2 dx_n \\ &= \frac{1}{2}x - \frac{1}{4}\sin 2x \Big|_0^{n\pi} \\ &= \frac{1}{2}n\pi \end{aligned}$$

$$\begin{aligned} \int_0^a u_0 j_0(x_n) x_n^2 dx_n &= \int_0^{n\pi} u_0 \left(\frac{\sin x_n}{x_n}\right) x_n^2 dx_n \\ &= u_0 (\sin x - x \cos x) \Big|_0^{n\pi} \\ &= (-1)^{n-1} u_0 n\pi \end{aligned}$$

于是：

$$A_n = \frac{1}{\|j_0(x_n)\|^2} \int_0^a u_0 j_0(x_n) x_n^2 dx_n = 2u_0 (-1)^{n-1}$$

原问题的解为：

$$\begin{aligned} u(r, t) &= 2u_0 \sum_{n=1}^{\infty} (-1)^{n-1} j_0\left(\frac{\mu_n r}{a}\right) e^{-\kappa\left(\frac{\mu_n}{a}\right)^2 t} \\ &= \frac{2u_0 a}{\pi r} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin\left(\frac{\mu_n r}{a}\right) e^{-\kappa\left(\frac{\mu_n}{a}\right)^2 t} \end{aligned}$$